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EXPERIMENT – 8

P***rivilege Escalation Using TryHackMe Platform***

Aim
To understand and demonstrate Linux privilege escalation techniques by exploiting system misconfigurations and vulnerabilities using the TryHackMe platform.

Tools / Platform Required

- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Linux Privilege Escalation Tools (LinPEAS, sudo, find, etc.)

Theory (Brief)
Privilege Escalation is a technique used to gain higher-level permissions than originally assigned.
In Linux systems, attackers often exploit misconfigured permissions, vulnerable binaries, or weak sudo rules to escalate from a normal user to root.

Steps to Perform the Experiment

Step 1: Login to TryHackMe
1. Openbrowserandgoto:https://tryhackme.com
2. Loginusingyourregisteredcredentials.

Step 2: Open Privilege Escalation Room

1. Inthesearchbar,type**LinuxPrivilegeEscalation**.
2. Clickontheappropriateroom.
3. Click**JoinRoom**.

Step 3: Start the AttackBox

1.Click **Start AttackBox**.
2. WaitfortheLinuxenvironmenttoload.

Step 4: Initial Enumeration

1. Check current user: whoami
2. Check system information: uname -a
3. Check sudo permissions: sudo -l

Step 5: Identify Privilege Escalation Vectors

1. Sudomisconfigurations
2. SUIDbinaries
3. Examplecommand:find/-perm-40002>/dev/null

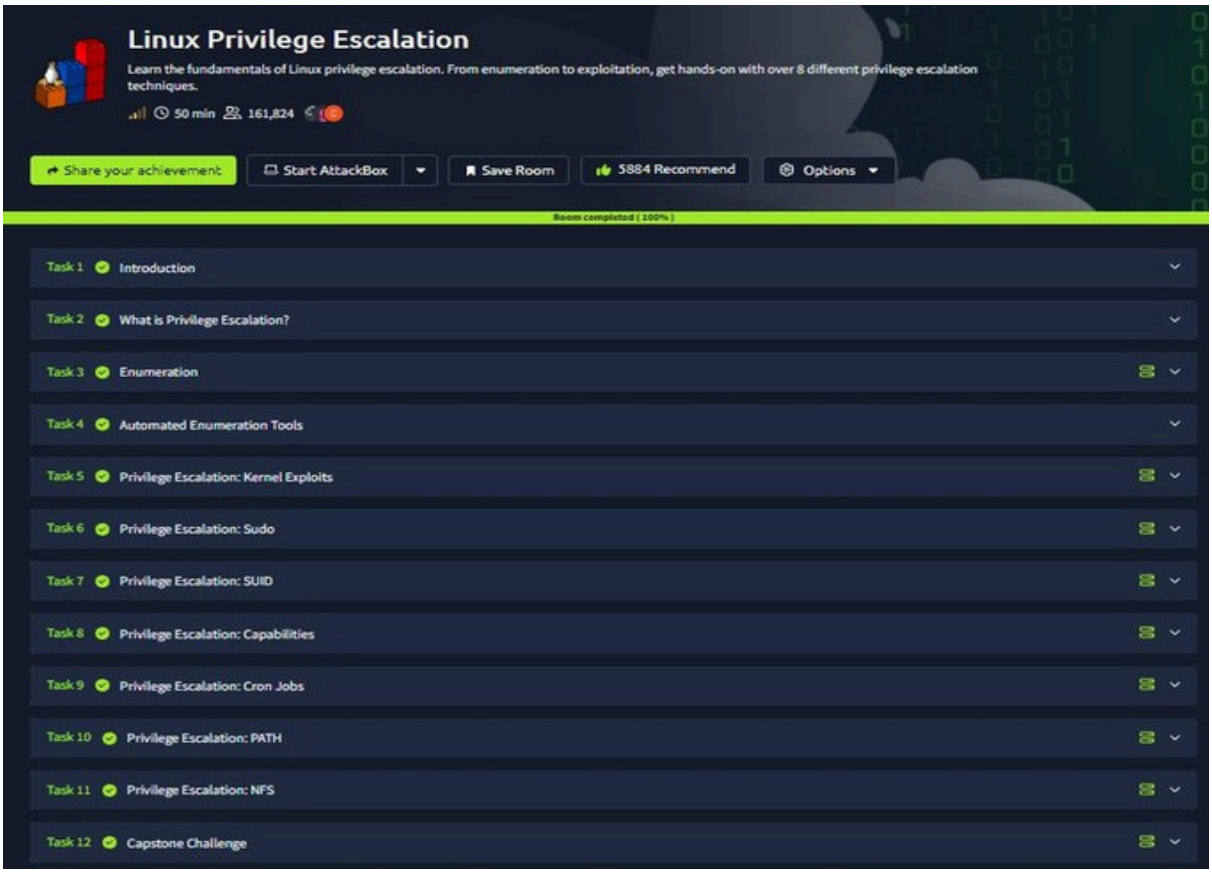
Step 6: Exploit Vulnerability

1. Executethevulnerablebinaryormisconfiguredcommand.
2. Escalateprivilegestorootuser.
3. Verifyescalation:whoami

Step 7: Capture Flags / Complete Tasks

1. Obtainroot-levelaccess.
2. Capturetherequiredflags.
3. Markalltasksascompleted.

Output



Result

Thus, privilege escalation was successfully performed using the TryHackMe platform. The experiment provided hands-on understanding of Linux privilege escalation techniques and system security weaknesses.